Journal of Disability Research

2025 | Volume 4 | Pages: 1–10 | e-location ID: e20240086
DOI: 10.57197/JDR-2024-0086

Smart Assistive Navigation System for Visually Impaired People

Gabriel Iluebe Okolo[1], Turke Althobaiti[2,*] and Naeem Ramzan[1]

[1]School of Computing, Engineering and Physical Sciences, University of the West of Scotland, Paisley, PA1 2BE, UK
[2]Department of Computer Science, Faculty of Science, Northern Border University, Arar, Saudi Arabia

Correspondence to:
Turke Althobaiti*, e-mail: Turke.althobaiti@nbu.edu.sa, Tel.: 00966146615844
Gabriel Iluebe Okolo, e-mail: gabriel.okolo@uws.ac.uk
Naeem Ramzan, e-mail: naeem.ramzan@uws.ac.uk

Received: April 21 2024; Revised: June 19 2024; Accepted: June 19 2024; Published Online: January 3 2025

ABSTRACT

It can be difficult for a visually impaired person to move on their own and use a typical white cane to identify items such as people, animals, crosswalks, pavements, and uneven terrain. A smart assistive navigation system is proposed in this research that combines voice-over with object detection. The system aims to promote the independence of visually impaired navigation by offering guidance with auditory feedback and tactile input to the visually impaired user upon object recognition. The created mobile application uses voice and audio only to provide navigational guidance to users. The object detection model used is YOLOV8, implemented on a Raspberry Pi equipped with a camera, speaker, ultrasonic sensor, and moisture sensor. The average accuracy score of nine tested obstacles using YOLOV8 was 91.70%. In addition to identifying objects, the built prototype allows the user to move between locations while giving them access to environmental data such as item distance, orientation, and surface wetness. The goal of this research is to enable people who are visually impaired to move about in both indoor and outdoor spaces securely and freely.

KEYWORDS

visually impaired person, artificial intelligence (AI), internet of things (IoT), object detection

INTRODUCTION

The World Health Organization estimates that at least 2.2 billion people worldwide are visually impaired (WHO, 2023). Visual impairment (VI) is the term used to describe a condition in which vision is limited and cannot be improved with medication or the use of vision correction lenses. There are several possible causes of VI. Delaying cataract surgery led to 272,000 visually impaired persons and 216,000 blind people, according to the Malaysian National Eye Survey II (Saud et al., 2021). Diabetes is the second most prevalent cause of blindness, accounting for 10% of instances of blindness and 6% of cases of impaired vision. Head traumas and other accidents are other factors contributing to these rising statistics. In order to address this human tragedy, academics are increasingly motivated to create novel developments in the field of assistive technology. One of the five senses that aids in our understanding of the outer world is sight, according to renowned philosopher and scientist Aristotle.

Early-life VI may result in poor performance in a variety of life domains. Apart from experiencing higher levels of anxiety or sadness, people who have VI frequently show lower levels of participation and productivity at work (Aymaz and Çavdar, 2016). Older adults may be more likely to fall due to VI. It also affects their ability to do routine duties. Furthermore, social isolation may occur, which may result in an early admission to a nursing home or other care facility. It is also difficult to move around freely and navigate in both new and familiar environments. Individuals seek guidance to ascertain the best course of action, the presence of solid things in their path, or any sign of moisture.

Many attempts have been made to create guards or obstacle detection systems for those with low vision. The most common solution is to use a basic walking aid that has sensors (Kalsoom et al., 2020) that can gather data about the surroundings. In the past, those who were blind or visually impaired used a wooden cane to go around, but over time, aluminum canes have overshadowed wooden canes. However as technology advances, the conventional walking stick may no longer be adequate to grant visually impaired people their desired freedom. Over time, several technological tools have been developed, such as the Mowat sensor, SensCap, and smart stick (Kher Chaitrali et al., 2015). The multisensory smart stick can identify impediments and deliver feedback

© 2025 The Author(s). This is an open access article distributed under the terms of the Creative Commons Attribution License (CC BY) 4.0, which permits unrestricted use, distribution and reproduction in any medium, provided the original author and source are credited.

to the user *via* a variety of modalities, including vibrations, audio alarms, and verbal prompts. Mobility aids (Dong, 2014) have historically been used by people with vision impairments and have a number of drawbacks. Some technologies need a separate power supply or a navigation system, which means that the user must carry it in a bag when they go outside (Patil et al., 2021). When faced with these large designs, the user is prone to become fatigued.

There are three main components commonly included in smart stick devices for visually impaired people: environmental sensing, which identifies obstacles and hazards in the immediate area; directional guidance, which shows which way to turn; and orientation support (Choong and Reddy, 2019). These functions are intended to support autonomous movement and improve users' comfort. There are certain restrictions associated with the traditional use of guide dogs and white canes by people who are visually impaired for mobility assistance. Despite being inexpensive, the white cane's ability to identify impediments is restricted because it only provides tactile feedback. As a result, there might be a smaller window of time for the person to react to the demands made by the scenario, which could be detrimental to their well-being. The purchase of a competent dog requires a significant outlay of cash.

Many academics are conducting in-depth studies to provide effective navigation support for those who are visually impaired. For those who are visually challenged, adding obstacle detection technology to a walking stick might improve their mobility. By providing audible notifications, users will be able to stay aware of their surroundings, which will greatly reduce the risk of accidents. To help people in confined spaces, an automatic voice-activated switching system has been included.

The main goal of this research is to create a smart stick that visually impaired people can use to become less dependent on others for mobility. It also aims to determine how effective object detection and monocular depth estimation are as components of a navigation system for the visually impaired. The ultrasonic sensor, Raspberry Pi, vibrator, moisture sensor, and camera module make up the recommended smart stick system. This paper proposes a "monocular depth estimation through ultrasonic sensor" approach that generates a depth image from a camera in order to solve the problem of assessing depth using only one Red, Green, Blue (RGB) camera. The research uses computer vision techniques like the you only look once (YOLO) algorithm to recognize things in real time in photos and videos; we use an ultrasonic sensor for monocular depth estimation to ascertain an object's depth and use a moisture sensor to detect the surface state. With the aid of a sensor, the system creates audio that changes in strength in accordance with the distance of the detected item, resulting in a sort of spatial audio and haptic impact. This project also includes creating, testing, and developing a prototype smartphone application that gives instructions depending on the visually impaired user's environment. This prototype uses voice instructions to guide the user provided by the Global Positioning System (GPS) navigation app, and in order to assist the user in getting where they are going, an object detection model detects items in front of them.

The contribution of this work can be summarized as follows: (i) we propose a light weight and portable smart stick for the visually impaired using artificial intelligence (AI)-based object detection and Internet of Things (IoT) sensor technology. (ii) We propose a smartphone navigation application for visually impaired people. (iii) We evaluate the performance of the YOLOV8 algorithm and the IoT sensors. (iv) We provide a detailed performance evaluation of the proposed smart assistive navigation system.

RELATED WORK

People who are visually impaired have several challenges in their day-to-day lives. They find it difficult to move freely and navigate both familiar and new settings. People ask for advice to determine whether they are going in an appropriate direction or if they are any obstructions in their path, and so on. There are several wearable gadgets available today that are intended to help those who are visually impaired (Divya et al., 2019). As part of their everyday routine, people with vision impairments may find it easier to navigate surroundings with the use of assistive technologies. Okolo et al. (2024) carried out a review that discusses assistive technologies designed to enhance the mobility and safety of visually impaired individuals. Hassan et al. (2021) proposed using a stick with sensors to help those who are visually impaired. The main goal of this design is to provide a tool that will allow people with VIs to recognize items in different orientations and ground-level apertures, including pits and manholes, in order to enable them to move around freely. The suggested approach makes use of several sensors with the capacity to recognize impediments in order to prevent collisions and to see things from different angles (Al-Muqbali et al., 2020). An extra sensor is positioned at the walking stick's lowest point with the purpose of detecting underground holes. The playback and audio recorder units were fitted with these sensors. A prototype was designed using Pro/E Creo 5.0 software and includes various hardware components such as "ATmega8" microcontroller, sensors, a power supply, a servomotor, a buzzer, equipment for capturing and replaying voice recordings, and a speaker.

Bai et al. (2017) developed an assistive technology prototype for visually impaired people to identify impediments around their surroundings in every direction. The system has capabilities that can identify barriers, pits, and manholes on the terrain's top and frontal surfaces. According to Sahoo et al. (2019), those who are visually impaired are provided with a unique cane in order to aid in simple navigation. The cane has a sensor that can identify the presence of water. The obstacle detection system makes use of ultrasonic sensors, which use ultrasonic waves. The sensor has the ability to identify obstacles and then send the data it has collected to the microcontroller. After processing the data, the microcontroller determines how close the obstacle is to the person.

The research conducted by Nivedita et al. (2019) shows that ultrasonic calculations of angle and distance are accurate, with estimates showing proportionate errors and variances that are contained within a certain range. Dang et al. (2016) developed an assistive technology for the visually impaired. The device vibrates when it senses footprints, damp surfaces, or an obstruction. It could also send out personalized audio messages *via* speakers or headphones. The subjects had an

impressive capacity to avoid the obstacles, as demonstrated by their very quick reaction time of 39 ms when the obstacles were placed at an average distance of 400 cm.

Paul et al. (2019) presented a portable gadget that combines an obstacle-detecting circuit with an infrared sensor to function on a stick. It was primarily designed to help those who are visually impaired navigate their environment safely and independently while avoiding possible hazards. In Chen et al. (2018) a single ultrasonic sensor was used to identify barriers in front of the body, while two more sensors were used to identify objects above and below the knee. Buzzers were used by the moisture sensor to provide feedback in order to identify puddles and alert the user. Their design was unique in that it included a radio frequency module for stick location tracking, consisting of a transmitter on the user and a receiver on the stick.

Ullah et al. (2018) proposed a voice-activated smart walking stick to help those who are visually impaired. The system consists of basic walking assistance with ultrasonic sensors that can identify water, holes, and obstructions. When GPS technology is combined with pre-programmed destinations, those who are visually impaired may travel ideal routes with ease. Furthermore, the activation of devices uses voice commands. The system is made up of two ultrasonic sensors—the Pit and Water sensors—as well as a GPS receiver, speech synthesizer, ATmega328/P microcontroller, speaker, battery, and Global System for Mobile Communications (GSM) module. Pieralisi et al. (2017) took an interesting method to developing a smart stick for the visually impaired. The researchers added the GPS and GSM modules, which allowed the user to text their carer their position coordinates. Moreover, an easy-to-use interface was designed to allow the user's guardians to change the message recipient's mobile number.

A system that uses vision to help people with VIs navigate was presented by Real and Araujo (2019). To help visually challenged people discover objects, this system has a camera, a haptic feedback device, and an integrated computer. Wearable technology helps those who are visually impaired move more easily. According to the research, this specific technology helps visually impaired people navigate by avoiding accidents with obstructions. According to Zhao et al. (2020), people with VIs may benefit from the use of an assistive technology that uses the optical character recognition algorithm to help them read. The device consists of a customized glove intended to help those who are visually impaired navigate their environment. To aid in this endeavor, the glove boasts an embedded camera index. With the help of a gloved index finger, the visually impaired person reads the first sentence from left to right. The camera that is positioned below the finger will process the textual picture and then provide an audio output.

Visually impaired people can now navigate outdoors more easily with the help of IoT-based applications and systems. BlinDar, which was put out by Saquib et al. (2017), is one such application.

By utilizing the Arduino integrated development environment and ultrasonic sensors coupled with a Wi-Fi module, Kunta et al. (2020) developed a smart stick utilizing IoT technologies. After seeing how the sensors functioned, they programmed the microcontroller to play the phrase "obstacle ahead" whenever an obstacle crosses the 100-cm barrier. Additionally, the smart stick can notify others when emergency aid is needed. Chava et al. (2021) developed an IoT-based smart shoe system. Microcontrollers equipped with sensors and buzzers have been utilized. The smart shoe uses buzzers to alert visually impaired users when it detects an issue. They have used integrated sensors and the IoT to create smart eyewear that increase the effectiveness of the smart shoe. Kumar et al. (2019a) proposed an IoT wearable navigation system for visually impaired people that uses the NEO-6M GPS module to determine the user's current location in real time. They noted a 90% accuracy rate. They made use of the Raspberry Pi's built-in text-to-speech feature for audio output, as well as an ultrasonic sensor and camera.

Rahman et al. (2021) propose the use of IoT and deep learning techniques to link a smart cap with a smart blind cane. Bluetooth- and Wi-Fi-connected deep learning techniques are used by the smart cap. Krishnan et al. (2021) presented an IoT-based navigation system that uses an Arduino, a vibration sensor, a piezo buzzer, and a push button to notify the user's exact position to their care home for urgent emergency assistance. This system recognizes the surface barriers in the immediate vicinity. Using voice instructions and a buzzer, the system sends out an alarm. GPS is used to determine the exact position. Pan et al. (2022) proposed an AI deep learning model for IoT connection. They have proposed that the system may be expanded to provide non-orthogonal connection for multiple-input multiple-output, holographic multiple input multiple output surfaces, and reconfigurable intelligent surfaces. They attained a classification accuracy of 99.

A study by Choudhary et al. (2020) uses an Radio Frequency Identification (RFID) model to schedule public transportation and assist visually impaired people in getting to bus stops by using GPS to provide audio output. They obtained the information on public transit by using two modules: the bus and user modules. Pathak et al. (2020) presented a method to identify stairs and impediments close to the visually impaired user that uses infrared and ultrasonic sensors. They have suggested a cheap, responsive navigation system with little power usage. Alarm feeds have been used to notify visually impaired people if an obstruction is detected. They used an Arduino microcontroller and ultrasonic sensors to identify impediments.

A theoretical model for navigation with electronic assistance was presented by Mala et al. (2017). They have employed a Bluetooth stick connected with a GPS-integrated stick and headphones. The model facilitates the vision handicapped in getting to where they are going. Ultrasonic sensors, a GPS module for navigation, a speech synthesizer, a GPS receiver, and a microcontroller for the stick were all utilized. The user's location is determined using GPS, and they can get audio instructions to lead them to their destination *via* a headset that is connected to Bluetooth.

An IoT-based solution with wearable glasses and walking assistance for visually impaired people was proposed by Zhou et al. (2017). The Alf and Vegard's RISC processor chip in the glasses is used to link the sensors. A portable GPS module is used to collect data from the glasses and walking stick, and a GPRS module is used to obtain position information *via* the cloud. The user receives voice commands to direct them using their smartphone's portable GPS module.

Rahman et al. (2018) created a stick for the visually impaired that uses an Arduino NANO microcontroller, sensors, and GSM to determine the user's location and relay information. They recommended the model since it was easily available, cost-effective, and had good object detection.

An image recognition and navigation system was presented by Kumar et al. (2019b) to detect impediments and give feedback to the visually impaired user *via* audio messages. When an impediment is detected by the ultrasonic sensor, the user is given a spoken order to move to the right or left. Additionally, they made use of a face recognition model that can assist visually impaired people in identifying the owner of their smartphone. Chaurasia et al. (2022) proposed an indoor autonomous navigation system. Three modules have been created for the system: an RFID-based destination detection module, an object identification module utilizing ultrasonic sensors, and a navigation module using a Raspberry Pi, ultrasonic sensor, and camera. When an item is detected, the model uses offline embedded text-to-speech technology to give voice command to the user, indicating which way to turn (blue light to navigate right) and (red light to navigate left).

A clever solution for visually impaired people was put up by Messaoudi et al. (2020), who gave them a smart white cane and sound buzzers to allow them to interact with their surroundings. They made use of accelerometers, cameras, and microcontrollers. The gadget alerts the visually impaired user when an item is spotted using buzzers.

A study on indoor navigation for the visually impaired using wearable audio aid was delivered by Vamsi Krishna and Aparna (2018). To enhance ultrasonic perception, they have employed two ultrasonic sensors, an RGB camera, and a Raspberry Pi. The user's alert messages are sent from cloud-stored data. They have found that, on average, users take 100 s to reach their destination while avoiding obstacles, with a 97.33% success rate. They employed a camera and wore glasses with ultrasonic sensors to detect obstacles.

According to Lopes et al. (2014), there are two use cases for the IoT: visually impaired individuals and neurological impaired individuals. They suggested a sensor-based application that would leverage Bluetooth and Wi-Fi wireless networks in conjunction with voice command inputs from visually impaired users. They proposed using triangulation to determine the user's present location and determine their position. A technique was presented by See et al. (2022) to assist visually impaired individuals in determining the location of obstacles, such as those affecting the head, torso, ground, or entire body. Within 1.6 m, the technology can identify obstructions. Gestures and voice commands can be used to operate the system.

A cost-effective and user-friendly assistance system using GPS, vibration motors, buzzers, and ultrasonic sensors was presented by Saud et al. (2021) for those who are visually impaired. They made use of the vibrating motors on the walking stick and ultrasonic sensors attached to a hat. The vibration motors vibrate in response to the detection of an obstruction. To direct the person to the spot, they used two switches on the walking stick to transmit signals to the phone. To guide the user, they created an application with an Massachusetts Institute of Technology app inventor.

METHODOLOGY

Smart stick application

The proposed system provides the user with vision. The block diagram comprises the Raspberry Pi 3, an ultrasonic sensor, a moisture sensor, an accelerometer/gyroscope, a camera, a sound output module, and a vibration module. An overview of the proposed architecture is provided in Figure 1. In the proposed method, an ultrasonic sensor is used to measure the distance between an impediment and the user. The sensors produce sound waves in front of the barrier that have an accurate distance detection capability. The sensor is positioned precisely to enable reliable data collection and processing.

The main controller for the system is the Raspberry Pi. The ultrasonic sensor on the Raspberry Pi allows for continuous obstacle distance measurement. The ultrasonic sensor

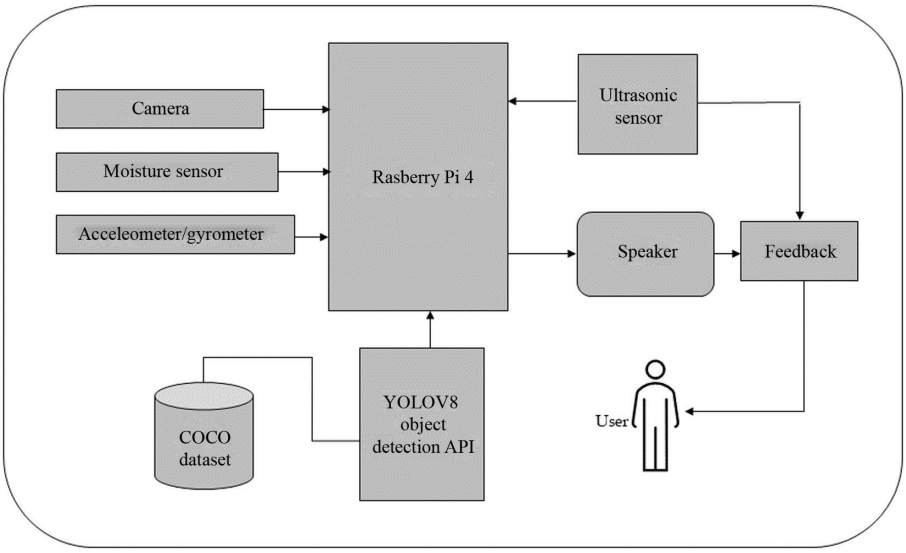

Figure 1: Overview diagram of sensors connected to the Raspberry Pi.

uses the time it takes for ultrasonic waves to travel round trip to and from the barrier to determine the distance. When an obstruction is found within 450 cm of the ultrasonic sensor, the Raspberry Pi receives a signal from the sensor. When the Raspberry Pi detects an impediment, it activates the web camera attached to it. When the camera is turned on, the photo is captured. This image is delivered concurrently to the Raspberry Pi, which accesses the COCO dataset consisting of several example images of different types of obstacles. The obstacle is analyzed and classified using YOLOV8. Upon detection, the sound module provides the user with the object's name as an output in voice. The moisture sensor is used to detect wet surfaces.

Raspberry Pi

The Raspberry Pi is the main controlling component of the system. Several inputs are sent to the Raspberry Pi board through the general purpose input/output pins, such as cameras, and ultrasonic sensors. This device is the brain of our system. It will handle all communication and data processing.

YOLOV8 object dectection model

The YOLOV8 model was introduced by Redmon et al. (2016) and is very good at object recognition and simple to use. The YOLO approach uses convolutional neural networks (CNNs) to detect things. The object detection approach involves two distinct tasks. The first objective is to locate the entity precisely; the second is to categorize such entities. It has been recommended to utilize a "region-based CNN" or one of its variants for object detection. Ren et al. (2016) describe a neural network for image processing that divides images into several regions, each assigned bounding boxes and corresponding probabilities. This network, known as YOLO or Darknet-53, is a complete CNN network consisting of 53 convolutional layers and using a stride for downsampling. For example, an input image size of 512 × 512 with a stride of 16 results in an output size of 32 × 32. YOLO is noted for its simple design, high computational efficiency, and support for end-to-end training, making it suitable for a wide range of object detection tasks. The COCO dataset, which comprises over 200,000 images of 80 item categories covering commonplace objects including trees, dogs, traffic signs, people, seats, crosswalks, and more, is used to train the object recognition YOLO model.

Camera module

When an object comes within the range of the camera sensor, it will use its capabilities to capture light entering the lens and convert it into electrical signals, which will be stored as digital data. After the data are stored, the next step is pre-processing, which involves removing noise, adjusting exposure, adjusting white balance, and improving other factors that will improve the quality of the image to be processed. Next, the camera processor function identifies the object's features, such as edges, shapes, texture, and enhancement of visibility, which will then be used by the YOLOV8 algorithm to identify the objects and assign labels to them.

Ultrasonic sensor

It is used to detect objects and for distance calculations. When it identifies a nearby obstacle, it notifies the user. The ultrasonic sensor uses high-frequency waves to accomplish its job. The item reflects those ultrasonic waves, which the ultrasonic sensor picks up on. The calculation of the distance is done using the following formula:

$$\text{Distance (cm)} = \frac{\text{Speed of sound (cm/}\mu\text{s)} \times \text{Time (}\mu\text{s)}}{2}$$

The controller plays an audio message through speakers or headphones and creates signals to indicate obstacles when it detects objects within a predetermined range. Using YOLO, the camera recognizes things and detects impediments.

Moisture/water sensor

A water sensor on the smart stick is meant to sense the presence of water when it is submerged in it. When the water sensor comes into contact with water, it shorts the circuit, resulting in a closed circuit that generates the desired output. The water sensor is helpful near any equipment that has the potential to leak water as well as in areas that are often used.

Vibration and sound output module

Vibration is utilized to transmit information and enhance the user experience. In the subject of haptic feedback, which is being developed for gadgets like phones, wearables, and gaming controllers, vibrators play a part in producing vibration. The system functions when the vibrator receives electric impulses that cause it to vibrate at a particular frequency and strength.

Accelerometer/gyroscope

The accelerometer/gyroscope detects the orientation of the device. These data offer insights into positioning the device and helping in navigation.

Smartphone navigation application

The smartphone navigation helps the vision impaired get about and go from one point to another. The Android application "Here We Go" is developed on Android Studio using Kotlin. Google Cloud console application programming interfaces are used for GPS navigation. Features like a map display, an e-mail button, and prompts for confirmation are all part of the user interface. This program, which combines real-time navigation, voice recognition, and emergency help, offers visually impaired people a hands-free option. By offering an accessible method of outside movement, this two-part smartphone application seeks to empower visually impaired people and ease their difficulties. The app is effective and user-friendly, as speech recognition, voice commands, and emergency features are included. For those who are visually impaired, real-time navigation improves the user experience and makes it a useful tool in their everyday lives. The Pseudocode for the Smartphone Navigation Application can be seen in Algorithm 1.

Algorithm 1: Pseudocode for smartphone navigation application

```
Result: Navigate to user-specified destination
Start;
Display "Welcome Message";
Check Location and Audio permissions;
if user clicks email button then
    Send Email Alert;
else
    Initiate Speech Recognition;
    User speaks Destination;
    App asks for confirmation message as "YES" or "RECORD";
    if user says "YES" then
        Start navigation;
        while user has not reached destination do
        end
    else
        if user says "RECORD" then
        end
    end
end
```

RESULTS AND DISCUSSION

We present the results obtained from the implementation of this project focusing on advanced object detection using the YOLOV8 framework paired with sensor technologies, as shown in Table 1. We used the COCO dataset, a sizable image recognition dataset designed for object identification, segmentation, and captioning model training and assessment, to confirm the findings. The photos cover a broad range of item categories and are taken in a variety of settings, including both indoor and outdoor situations. The excellent annotations in the COCO dataset provide comprehensive details on the objects in each image. For example, every object has a label that describes the precise bounds of the object in the image, together with the category and bounding box coordinates. Real-time object detection scenarios are provided in Figure 2. The smartphone navigation prototype, as shown in Figure 3, illustrates how the app is used. A user simply needs to say where he/she is going, and then a map will be open which uses voice recognition to direct the user to their location.

Table 1: Performance analysis on objects for both indoor and outdoor setting.

Obstacles	Accuracy	Detection time	Distance accuracy
Person	92.40	0.001 s	True
Car	92.00	0.001 s	True
Bus	90.98	0.4 s	True
Truck	88.89	0.2 s	True
Chair	94.00	0.2 s	True
TV	91.00	0.3 s	True
Couch	93.00	0.2 s	True
Bottle	90.00	0.1 s	True
Refrigerator	93.00	0.1 s	True

Test results

The system detects a person with an accuracy of 92.40% with a low detection time of 0.001 s, indicating an efficient and reliable detection score. Cars showed a high accuracy of 92.00%, also with a detection time of 0.001 s, suggesting the system's capability to promptly and accurately identify moving vehicles. Buses attained an accuracy of 90.98% and a longer detection time of 0.4 s, reflecting a trade-off between detecting larger objects and the time required for processing. Trucks had an accuracy of 88.89% and a detection time of 0.2 s, which is reasonable given their size. Chairs are detected with an accuracy of 94.00% and a detection time of 0.2 s, showcasing the system's proficiency in recognizing common indoor obstacles. TV detection accuracy was 91.00% with a detection time of 0.3 s. Couches had an accuracy of 93.00% and the same detection time of 0.2 s as chairs, suggesting a consistent performance for indoor furniture. Bottles attained an accuracy of 90.00% with a detection time of 0.1 s. Finally, the refrigerator was detected with an accuracy of 93.00% and a detection time of 0.1 s, reflecting efficiency in identifying large, stationary indoor objects. Across all categories, the distance accuracy is consistently true, indicating reliable distance estimation by the system.

This experiment's validation is based on different factors, such as the high accuracy rates across most categories, which suggests that the system is robust and effective in identifying various obstacles and in measuring their distance. The detection times, particularly the extremely low times for people and cars, indicate that the system is capable of real-time detection, which is crucial for the safety of visually impaired users. The consistent distance accuracy across all obstacles further supports the system's reliability in providing accurate spatial information, which is essential for navigation. A representation of the object detection and distance estimation performance analysis is provided in Figure 4.

Discussion

The integration of a camera for object detection enhances the system's ability to recognize a wide variety of objects. This visual input provides information about the environment by detecting objects, which is crucial for navigation. The use of an ultrasonic sensor to measure the distance to objects provides accurate, real-time data about the user's immediate surroundings. This helps in preventing collisions and navigating around obstacles effectively. Combining camera-based object detection with ultrasonic distance measurement leverages the strengths of both technologies. While the camera can identify and classify objects, the ultrasonic sensor provides precise distance measurements, making the navigation system more reliable and comprehensive.

The integration of camera and ultrasonic sensors represents a promising approach to developing more effective navigation aids for visually impaired individuals. By combining visual and distance information, the system can offer more comprehensive assistance than traditional

Figure 2: Obstacle detection using the YOLOV8 model.

single-sensor systems. Such systems can greatly enhance the independence and mobility of visually impaired users, allowing them to navigate complex environments more safely and confidently.

The research highlights the potential for sensor integration and data processing within assistive devices. This could pave the way for the development of more advanced, multifunctional assistive technologies. The findings suggest opportunities for creating more personalized and customizable assistive solutions. By refining sensor fusion and feedback mechanisms, devices can be tailored to meet the specific needs and preferences of individual users.

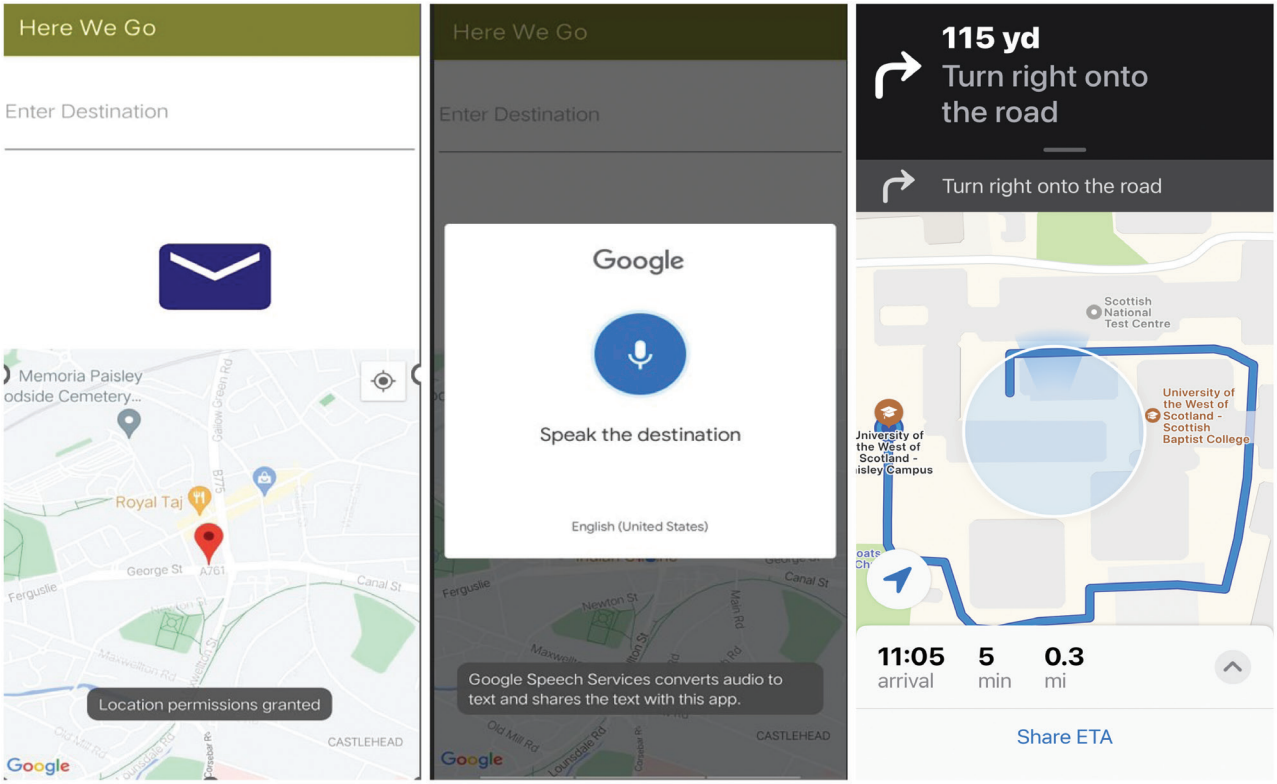

Figure 3: Smartphone navigation application.

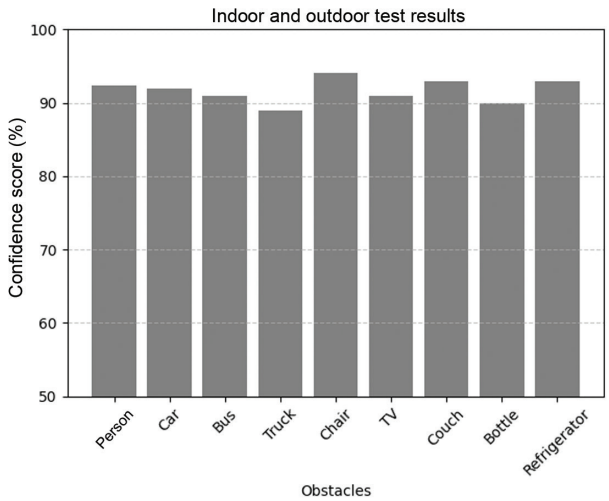

Figure 4: Results for indoor and outdoor obstacle detection using the YOLOV8 model.

CONCLUSIONS

The overall system proved to be effective in assisting the visually impaired. The obstacle detection algorithm integrated with ultrasonic and moisture sensors has yielded remarkably accurate distance measurements across all tested objects. Overall, the test results are indicative of a robust obstacle detection system with strengths in identifying obstacles and distance measurement. The ultrasonic sensor is used to identify obstacles in the user's path and provide haptic feedback *via* the vibrator. A moisture sensor identifies rain or other damp conditions, warning the user of potentially hazardous conditions or slippery surfaces. The addition of a camera to the stick for object detection allows the user to move and understand their environment better by providing additional contextual information. The use of a smartphone navigation app also helps to improve the overall experience for the visually impaired by directing them to their destination *via* audio. Because of the integration of various technologies, users can now navigate complex environments with ease. Their improved flexibility and movement may allow them to engage with their surroundings more and explore opportunities that were previously unattainable. Through this technology, individuals with VIs can participate more completely in daily activities and overcome hurdles, promoting greater accessibility and participation in society.

Further development will concentrate on increasing the system's capacity to adjust to a variety of unanticipated situations. To do this, it will be necessary to add more sensors and enhance the system's learning algorithms to better distinguish between obstacles with similar features.

AUTHOR CONTRIBUTIONS

GIO conceptualized the study and took care of methodology, software, validation, data curation, writing—original

draft, writing—review and editing, visualization, and supervision. TA conceptualized the study and took care of methodology, validation, and visualization. NR conceptualized the study and took care of methodology, writing—original draft, writing—review and editing, supervision, and project administration.

ACKNOWLEDGMENTS

The authors extend their appreciation to the King Salman Center for Disability Research for funding this work through Research Group no. KSRG-2023-374 (funder ID: http://dx.doi.org/10.13039/501100019345).

REFERENCES

Al-Muqbali F., Al-Tourshi N., Al-Kiyumi K. and Hajmohideen F. (2020). Smart technologies for visually impaired: assisting and conquering infirmity of blind people using AI technologies. In: *Proceedings of the 2020 12th Annual Undergraduate Research Conference on Applied Computing (URC)*, IEEE, Piscataway, NJ; pp. 1-4.

Aymaz S. and Çavdar T. (2016). Ultrasonic assistive headset for visually impaired people. In: *Proceedings of the 2016 39th International Conference on Telecommunications and Signal Processing (TSP)*, IEEE, Piscataway, NJ; pp. 388-391.

Bai J., Lian S., Liu Z., Wang K. and Liu D. (2017). Smart guiding glasses for visually impaired people in indoor environment. *IEEE Trans. Consum. Electron.*, 63, 258-266.

Chaurasia M.A., Rasool S., Afroze M., Jalal S.A., Zareen R., Fatima U., et al. (2022). Automated navigation system with indoor assistance for blind. In: *Contactless Healthcare Facilitation and Commodity Delivery Management During COVID 19 Pandemic*, Springer Nature, Berlin; pp.119-128.

Chava T., Srinivas A.T., Sai A.L. and Rachapudi V. (2021). IoT based smart shoe for the blind. In: *Proceedings of the 2021 6th International Conference on Inventive Computation Technologies (ICICT)*, IEEE, Piscataway, NJ; pp. 220-223.

Chen J., Xu W., Peng W., Bu W., Xing B. and Liu G. (2018). Road object detection using a disparity-based fusion model. *IEEE Access*, 6, 19654-19663.

Choong L.T. and Reddy M.V. (2019). Smart assisting visually impaired stick (savis). *Int. J. Eng. Res. Technol.*, 8, 78-84.

Choudhary S., Bhatia V. and Ramkumar K. (2020). IoT based navigation system for visually impaired people. In: *Proceedings of the 2020 8th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)*, IEEE, Piscataway, NJ; pp. 521-525.

Dang Q.K., Chee Y., Pham D.D. and Suh Y.S. (2016). A virtual blind cane using a line laser-based vision system and an inertial measurement unit. *Sensors*, 16, 95.

Divya S., Raj S., Shai M.P., Akash A.J. and Nisha V. (2019). Smart assistance navigational system for visually impaired individuals. In: *Proceedings of the 2019 IEEE International Conference on Intelligent Techniques in Control, Optimization and Signal Processing (INCOS)*, IEEE, Piscataway, NJ; pp. 1-5.

Dong H. (2014). *Indoor Navigation System for the Visually Impaired with User-Centric Graph Representation and Vision Detection Assistance*. Thesis, University of Massachusetts Amherst, Amherst, MA.

Hassan M.K., Hassan M.R., Ahmed M.T., Sabbir M.S.A., Ahmed M.S. and Biswas M. (2021). A survey on an intelligent system for persons with visual disabilities. *Aust. J. Eng. Innov. Technol.*, 3, 97-118.

Kalsoom T., Ramzan N., Ahmed S. and Ur-Rehman M. (2020). Advances in sensor technologies in the era of smart factory and industry 4.0. *Sensors*, 20, 6783.

Kher Chaitrali S., Dabhade Yogita A., Kadam Snehal K., Dhamdhere Swati D. and Deshpande Aarti V. (2015). An intelligent walking stick for the blind. *Int. J. Eng. Res. Gen. Sci.*, 3, 1057-1062.

Krishnan R.S., Narayanan K.L., Murali S.M., Sangeetha A., Ram C.R.S. and Robinson Y.H. (2021). IoT based blind people monitoring system for visually impaired care homes. In: *Proceedings of the 2021 5th International Conference on Trends in Electronics and Informatics (ICOEI)*, IEEE, Piscataway, NJ; pp. 505-509.

Kumar N.A., Thangal Y.H. and Beevi K.S. (2019a). IoT enabled navigation system for blind. In: *Proceedings of the 2019 IEEE R10 Humanitarian Technology Conference (R10-HTC) (47129)*, IEEE, Piscataway, NJ; pp. 186-189.

Kumar P.M., Gandhi U., Varatharajan R., Manogaran G., Jidhesh R. and Vadivel T. (2019b). Intelligent face recognition and navigation system using neural learning for smart security in internet of things. *Cluster Comput.*, 22, 7733-7744. 10.1007/s10586-017-1323-4.

Kunta V., Tuniki C. and Sairam U. (2020). Multi-functional blind stick for visually impaired people. In: *Proceedings of the 2020 5th International Conference on Communication and Electronics Systems (ICCES)*, IEEE, Piscataway, NJ; pp. 895-899.

Lopes N.V., Pinto F., Furtado P. and Silva J. (2014). IoT architecture proposal for disabled people. In: *Proceedings of the 2014 IEEE 10th International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob)*, IEEE, Piscataway, NJ; pp. 152-158.

Mala N.S., Thushara S.S. and Subbiah S. (2017). Navigation gadget for visually impaired based on IoT. In: *Proceedings of the 2017 2nd International Conference on Computing and Communications Technologies (ICCCT)*, IEEE, Piscataway, NJ; pp. 334-338.

Messaoudi M.D., Menelas B.A.J. and Mcheick H. (2020). Autonomous smart white cane navigation system for indoor usage. *Technologies*, 8, 37.

Nivedita A., Sindhuja M., Asha G., Subasree R. and Monisha S. (2019). Smart cane navigation for visually impaired. *Int. J. Innovat. Technol. Explor. Eng.*, 8(6), 190-192.

Okolo G.I., Althobaiti T. and Ramzan N. (2024). Assistive systems for visually impaired persons: challenges and opportunities for navigation assistance. *Sensors*, 24, 3572.

Pan J., Ye N., Yu H., Hong T., Al-Rubaye S., Mumtaz S., et al. (2022). AI-driven blind signature classification for IoT connectivity: a deep learning approach. *IEEE Trans. Wireless Commun.*, 21, 6033-6047.

Pathak A., Adil M., Rafa T.S., Ferdoush J. and Mahmud A. (2020). An IoT based voice controlled blind stick to guide blind people. *Int. J. Eng. Invent.*, 9, 9-14.

Patil K., Kharat A., Chaudhary P., Bidgar S. and Gavhane R. (2021). Guidance system for visually impaired people. In: *Proceedings of the 2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS)*, IEEE, Piscataway, NJ; pp. 988-993.

Paul I.J.L., Sasirekha S., Mohanavalli S., Jayashree C., Priya P.M. and Monika K. (2019). Smart eye for visually impaired-an aid to help the blind people. In: *Proceedings of the 2019 International Conference on Computational Intelligence in Data Science (ICCIDS)*, IEEE, Piscataway, NJ; pp. 1-5.

Pieralisi M., Di Mattia V., Petrini V., De Leo A., Manfredi G., Russo P., et al. (2017). An electromagnetic sensor for the autonomous running of visually impaired and blind athletes (part II: The wearable device). *Sensors*, 17, 381.

Rahman M.W., Islam R. and Harun-Ar-Rashid M. (2018). IoT based blind person's stick. *Int. J. Comput. Appl.*, 182, 19-21.

Rahman M.W., Tashfia S.S., Islam R., Hasan M.M., Sultan S.I., Mia S., et al. (2021). The architectural design of smart blind assistant using IoT with deep learning paradigm. *Int. Things* 13, 100344.

Real S. and Araujo A. (2019). Navigation systems for the blind and visually impaired: past work, challenges, and open problems. *Sensors*, 19, 3404.

Redmon J., Divvala S., Girshick R. and Farhadi A. (2016). You only look once: unified, real-time object detection. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, IEEE, Piscataway, NJ; pp. 779-788.

Ren S., He K., Girshick R. and Sun J. (2016). Faster r-cnn: Towards real-time object detection with region proposal networks. *IEEE Trans Pattern Anal Mach Intell.*, 39, 1137-1149.

Sahoo N., Lin H.W. and Chang Y.H. (2019). Design and implementation of a walking stick aid for visually challenged people. *Sensors*, 19, 130.

Saquib Z., Murari V. and Bhargav S.N. (2017). Blindar: an invisible eye for the blind people making life easy for the blind with internet of things (IoT). In: *Proceedings of the 2017 2nd IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT)*, IEEE, Piscataway, NJ; pp. 71-75.

Saud S.N., Raya L., Abdullah M.I. and Isa M.Z.A. (2021). Smart navigation aids for blind and vision impairment people. In: *Proceedings of the International Conference on Computational Intelligence in Information System*, Springer, Berlin; pp. 54-62.

See A.R., Sasing B.G. and Advincula W.D. (2022). A smartphone-based mobility assistant using depth imaging for visually impaired and blind. *Appl. Sci.*, 12, 2802.

Ullah S., Ahmed S., Ahmad W. and Jan L. (2018). Sensor based smart stick for blind person navigation. *Int. J. Sci. Eng. Res.*, 9.

Vamsi Krishna B. and Aparna K. (2018). IoT-based indoor navigation wearable system for blind people. In: *Artificial Intelligence and Evolutionary Computations in Engineering Systems: Proceedings of ICAIECES 2017*, Springer, Berlin; pp. 413-421.

WHO. (2023). *Blindness and Vision Impairment*. https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment. Accessed January 9, 2024.

Zhao Y., Kupferstein E., Rojnirun H., Findlater L. and Azenkot S. (2020). The effectiveness of visual and audio wayfinding guidance on smartglasses for people with low vision. In: *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, ACM, New York, NY; pp. 1-14.

Zhou M., Li W. and Zhou B. (2017). An IoT system design for blind. In: *Proceedings of the 2017 14th Web Information Systems and Applications Conference (WISA)*, IEEE, Piscataway, NJ; pp. 90-92.